Department of Applied Mathematics, University of Dhaka
Assignment 05

Second Year B S (Honors) 2025–2026

Course Title: Math Lab II Course Code: AMTH 250

Name: **Roll No.:** **Group:**

Write a *FORTRAN* program to solve each of the following problems. Always use files named according to the assignment number and problem number to take input and show output. For example, for Problem Y of Assignment X, use `in_aXqY.txt` and `out_aXqY.txt`.

1 A mechanical component undergoes damped oscillatory motion. Its displacement is $x(t) = 8\,exp(-\frac{t}{4})\ cos(2t)$, where x is in centimeters and t is in seconds. The velocity is $v(t) = \frac{dx}{dt}$. Generate the necessary displacement values and approximate the velocity at $t = 2\,s$ using:

(i) Three-point midpoint formula, $h = 0.25$.

(ii) Three-point endpoint formula, $h = 0.25$.

(iii) Five-point midpoint formula, $h = 0.50$.

(iv) Five-point endpoint formula, $h = 0.50$.

Tabulate the method, step size, approximate velocity, exact velocity, and absolute error.

2 The charge on a capacitor varies as $q(t) = 40\,(1 - exp\left(-\frac{t}{2}\right))$, where q is in millicoulombs and t in seconds. The current is $i(t) = \frac{dq}{dt}$.

Using Richardson extrapolation, approximate the current at $t = 4\,s$ to four decimal places. Begin with $h = 0.4$ and successively halve h. Display the extrapolation table.

3 A vehicle moves along a straight road. Its displacement $s\ (m)$ is recorded at times $t\ (s)$:

t (s)	0.0	0.2	0.4	0.6	0.8	1.0	1.2
s (m)	0.00	1.18	2.72	4.68	7.04	9.80	12.96

Use suitable numerical differentiation formulas to find velocity and acceleration at $t = 0\,s$ and $t = 1.2\,s$.

4 The heat-transfer rate is $H(t) = \frac{120t}{1+t^2}\ J/s$. Find the total heat transferred during the first $2.4\ s\ is:\ Q = \int_0^{2.4} \frac{120t}{1+t^2}\ dt$.

(a) Composite trapezoidal rule, 30 equal subintervals.

(b) Composite Simpson's 1/3 rule, 30 equal subintervals.

(c) Composite Simpson's 3/8 rule, 30 equal subintervals.

(d) Composite Weddle's rule, 30 equal subintervals.

The exact value is $Q = 60\ ln(6.76)$. Compare the approximations in a table including absolute and percentage errors.

5 The instantaneous power of an electrical device is $P(t) = t^2\ cos(t)$ watts. Its energy consumption from $t = 0$ to $t = \frac{\pi}{4}$ is: $E = \int_0^{\frac{\pi}{4}} t^2\ cos(t)\ dt$.

Write a FORTRAN program using Romberg integration to approximate E correctly to five decimal places. Display the complete Romberg extrapolation table and compare with the exact value $E = \sqrt{2}(\frac{\pi^2}{32} + \frac{\pi}{4} - 1)$.

N.B. Students should bring the required algorithms for their convenience.